SG No. 222 O_{h}^{2} $Pn\bar{3}n$ [cubic]

* plus set: +[0, 0, 0]

Table 1: Wyckoff site: 2a, site symmetry: 432

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48]

Table 2: Wyckoff site: 6b, site symmetry: 42.2

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[1,2,3,4,17,18,19,20]
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[5,6,7,8,13,14,15,16]
3	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[9,10,11,12,21,22,23,24]
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[25,26,27,28,41,42,43,44]
5	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[29,30,31,32,37,38,39,40]
6	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[33,34,35,36,45,46,47,48]

Table 3: Wyckoff site: 8c, site symmetry: .-3.

No.	position	mapping
1	[0, 0, 0]	[1,5,9,25,29,33]
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[2,7,12,26,31,36]
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[3,8,10,27,32,34]
4	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[4,6,11,28,30,35]
5	$\left[0, 0, \frac{1}{2}\right]$	[13,17,21,37,41,45]
6	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[14,19,24,38,43,48]
7	$\left[0, \frac{1}{2}, 0\right]$	[15,20,22,39,44,46]
8	$\left[\frac{1}{2}, 0, 0\right]$	[16,18,23,40,42,47]

Table 4: Wyckoff site: 12d, site symmetry: -4..

No.	position	mapping
1	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	[1,4,41,44]
2	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[2,3,42,43]
3	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	[5,8,37,40]
4	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	[6,7,38,39]
5	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	[9,12,46,47]
6	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[10,11,45,48]
7	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	[13,16,29,32]
8	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	[14,15,30,31]
9	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	[17,20,25,28]
10	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[18,19,26,27]
11	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[21,24,34,35]
12	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	[22,23,33,36]

Table 5: Wyckoff site: `12e`, site symmetry: `4..`

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,4,17,20]`
2	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[2,3,18,19]`
3	$\left[\frac{1}{4},\ x,\ \frac{1}{4}\right]$	`[5,8,13,16]`
4	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	`[6,7,14,15]`
5	$\left[\frac{1}{4},\ \frac{1}{4},\ x\right]$	`[9,12,22,23]`
6	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}-x\right]$	`[10,11,21,24]`
7	$\left[-x,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[25,28,41,44]`
8	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[26,27,42,43]`
9	$\left[\frac{3}{4},\ -x,\ \frac{3}{4}\right]$	`[29,32,37,40]`
10	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	`[30,31,38,39]`
11	$\left[\frac{3}{4},\ \frac{3}{4},\ -x\right]$	`[33,36,46,47]`
12	$\left[\frac{3}{4},\ \frac{3}{4},\ x+\frac{1}{2}\right]$	`[34,35,45,48]`

Table 6: Wyckoff site: `16f`, site symmetry: `.3.`

No.	position	mapping
1	$[x,\ x,\ x]$	`[1,5,9]`
2	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ x\right]$	`[2,7,12]`
3	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{2}-x\right]$	`[3,8,10]`
4	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	`[4,6,11]`
5	$\left[x,\ x,\ \frac{1}{2}-x\right]$	`[13,17,21]`
6	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	`[14,19,24]`
7	$\left[x,\ \frac{1}{2}-x,\ x\right]$	`[15,20,22]`
8	$\left[\frac{1}{2}-x,\ x,\ x\right]$	`[16,18,23]`
9	$[-x,\ -x,\ -x]$	`[25,29,33]`
10	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ -x\right]$	`[26,31,36]`
11	$\left[x+\frac{1}{2},\ -x,\ x+\frac{1}{2}\right]$	`[27,32,34]`
12	$\left[-x,\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	`[28,30,35]`
13	$\left[-x,\ -x,\ x+\frac{1}{2}\right]$	`[37,41,45]`
14	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	`[38,43,48]`
15	$\left[-x,\ x+\frac{1}{2},\ -x\right]$	`[39,44,46]`
16	$\left[x+\frac{1}{2},\ -x,\ -x\right]$	`[40,42,47]`

Table 7: Wyckoff site: `24g`, site symmetry: `2..`

No.	position	mapping
1	$\left[x,\ \frac{3}{4},\ \frac{1}{4}\right]$	`[1,4]`
2	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	`[2,3]`
3	$\left[\frac{1}{4},\ x,\ \frac{3}{4}\right]$	`[5,8]`
4	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[6,7]`
5	$\left[\frac{3}{4},\ \frac{1}{4},\ x\right]$	`[9,12]`
6	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-x\right]$	`[10,11]`
7	$\left[\frac{3}{4},\ x,\ \frac{1}{4}\right]$	`[13,16]`
8	$\left[\frac{3}{4},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	`[14,15]`

continued ...

Table 7

No.	position	mapping
9	$\left[x,\ \frac{1}{4},\ \frac{3}{4}\right]$	[17,20]
10	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	[18,19]
11	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}-x\right]$	[21,24]
12	$\left[\frac{1}{4},\ \frac{3}{4},\ x\right]$	[22,23]
13	$\left[-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	[25,28]
14	$\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{3}{4}\right]$	[26,27]
15	$\left[\frac{3}{4},\ -x,\ \frac{1}{4}\right]$	[29,32]
16	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[30,31]
17	$\left[\frac{1}{4},\ \frac{3}{4},\ -x\right]$	[33,36]
18	$\left[\frac{1}{4},\ \frac{3}{4},\ x+\frac{1}{2}\right]$	[34,35]
19	$\left[\frac{1}{4},\ -x,\ \frac{3}{4}\right]$	[37,40]
20	$\left[\frac{1}{4},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[38,39]
21	$\left[-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	[41,44]
22	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{4}\right]$	[42,43]
23	$\left[\frac{3}{4},\ \frac{1}{4},\ x+\frac{1}{2}\right]$	[45,48]
24	$\left[\frac{3}{4},\ \frac{1}{4},\ -x\right]$	[46,47]

Table 8: Wyckoff site: 24h, site symmetry: ..2

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ y\right]$	[1,18]
2	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ y\right]$	[2,20]
3	$\left[\frac{1}{4},\ y,\ \frac{1}{2}-y\right]$	[3,17]
4	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ \frac{1}{2}-y\right]$	[4,19]
5	$\left[y,\ \frac{1}{4},\ y\right]$	[5,15]
6	$\left[y,\ \frac{1}{4},\ \frac{1}{2}-y\right]$	[6,13]
7	$\left[\frac{1}{2}-y,\ \frac{1}{4},\ y\right]$	[7,16]
8	$\left[\frac{1}{2}-y,\ \frac{1}{4},\ \frac{1}{2}-y\right]$	[8,14]
9	$\left[y,\ y,\ \frac{1}{4}\right]$	[9,21]
10	$\left[\frac{1}{2}-y,\ y,\ \frac{1}{4}\right]$	[10,23]
11	$\left[y,\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	[11,22]
12	$\left[\frac{1}{2}-y,\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	[12,24]
13	$\left[\frac{3}{4},\ -y,\ -y\right]$	[25,42]
14	$\left[\frac{3}{4},\ y+\frac{1}{2},\ -y\right]$	[26,44]
15	$\left[\frac{3}{4},\ -y,\ y+\frac{1}{2}\right]$	[27,41]
16	$\left[\frac{3}{4},\ y+\frac{1}{2},\ y+\frac{1}{2}\right]$	[28,43]
17	$\left[-y,\ \frac{3}{4},\ -y\right]$	[29,39]
18	$\left[-y,\ \frac{3}{4},\ y+\frac{1}{2}\right]$	[30,37]
19	$\left[y+\frac{1}{2},\ \frac{3}{4},\ -y\right]$	[31,40]
20	$\left[y+\frac{1}{2},\ \frac{3}{4},\ y+\frac{1}{2}\right]$	[32,38]
21	$\left[-y,\ -y,\ \frac{3}{4}\right]$	[33,45]
22	$\left[y+\frac{1}{2},\ -y,\ \frac{3}{4}\right]$	[34,47]
23	$\left[-y,\ y+\frac{1}{2},\ \frac{3}{4}\right]$	[35,46]
24	$\left[y+\frac{1}{2},\ y+\frac{1}{2},\ \frac{3}{4}\right]$	[36,48]

Table 9: Wyckoff site: `48i`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	`[2]`
3	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}-z\right]$	`[3]`
4	$\left[x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	`[4]`
5	$[z,\ x,\ y]$	`[5]`
6	$\left[z,\ \frac{1}{2}-x,\ \frac{1}{2}-y\right]$	`[6]`
7	$\left[\frac{1}{2}-z,\ \frac{1}{2}-x,\ y\right]$	`[7]`
8	$\left[\frac{1}{2}-z,\ x,\ \frac{1}{2}-y\right]$	`[8]`
9	$[y,\ z,\ x]$	`[9]`
10	$\left[\frac{1}{2}-y,\ z,\ \frac{1}{2}-x\right]$	`[10]`
11	$\left[y,\ \frac{1}{2}-z,\ \frac{1}{2}-x\right]$	`[11]`
12	$\left[\frac{1}{2}-y,\ \frac{1}{2}-z,\ x\right]$	`[12]`
13	$\left[y,\ x,\ \frac{1}{2}-z\right]$	`[13]`
14	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	`[14]`
15	$\left[y,\ \frac{1}{2}-x,\ z\right]$	`[15]`
16	$\left[\frac{1}{2}-y,\ x,\ z\right]$	`[16]`
17	$\left[x,\ z,\ \frac{1}{2}-y\right]$	`[17]`
18	$\left[\frac{1}{2}-x,\ z,\ y\right]$	`[18]`
19	$\left[\frac{1}{2}-x,\ \frac{1}{2}-z,\ \frac{1}{2}-y\right]$	`[19]`
20	$\left[x,\ \frac{1}{2}-z,\ y\right]$	`[20]`
21	$\left[z,\ y,\ \frac{1}{2}-x\right]$	`[21]`
22	$\left[z,\ \frac{1}{2}-y,\ x\right]$	`[22]`
23	$\left[\frac{1}{2}-z,\ y,\ x\right]$	`[23]`
24	$\left[\frac{1}{2}-z,\ \frac{1}{2}-y,\ \frac{1}{2}-x\right]$	`[24]`
25	$[-x,\ -y,\ -z]$	`[25]`
26	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	`[26]`
27	$\left[x+\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$	`[27]`
28	$\left[-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[28]`
29	$[-z,\ -x,\ -y]$	`[29]`
30	$\left[-z,\ x+\frac{1}{2},\ y+\frac{1}{2}\right]$	`[30]`
31	$\left[z+\frac{1}{2},\ x+\frac{1}{2},\ -y\right]$	`[31]`
32	$\left[z+\frac{1}{2},\ -x,\ y+\frac{1}{2}\right]$	`[32]`
33	$[-y,\ -z,\ -x]$	`[33]`
34	$\left[y+\frac{1}{2},\ -z,\ x+\frac{1}{2}\right]$	`[34]`
35	$\left[-y,\ z+\frac{1}{2},\ x+\frac{1}{2}\right]$	`[35]`
36	$\left[y+\frac{1}{2},\ z+\frac{1}{2},\ -x\right]$	`[36]`
37	$\left[-y,\ -x,\ z+\frac{1}{2}\right]$	`[37]`
38	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[38]`
39	$\left[-y,\ x+\frac{1}{2},\ -z\right]$	`[39]`
40	$\left[y+\frac{1}{2},\ -x,\ -z\right]$	`[40]`
41	$\left[-x,\ -z,\ y+\frac{1}{2}\right]$	`[41]`
42	$\left[x+\frac{1}{2},\ -z,\ -y\right]$	`[42]`
43	$\left[x+\frac{1}{2},\ z+\frac{1}{2},\ y+\frac{1}{2}\right]$	`[43]`
44	$\left[-x,\ z+\frac{1}{2},\ -y\right]$	`[44]`
45	$\left[-z,\ -y,\ x+\frac{1}{2}\right]$	`[45]`
46	$\left[-z,\ y+\frac{1}{2},\ -x\right]$	`[46]`

continued ...

Table 9

No.	position	mapping
47	$\left[z+\frac{1}{2}, -y, -x\right]$	[47]
48	$\left[z+\frac{1}{2}, y+\frac{1}{2}, x+\frac{1}{2}\right]$	[48]